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Toward Device-Free Micro-Gesture Tracking via Accurate Acoustic Doppler-Shift Detection

WENYUAN LIU¹, WEIHANG SHEN¹, BINBIN LI^{1,2}, AND LIN WANG¹

¹School of Information Science and Engineering, Yanshan University, Qinhuangdao 066004, China

²School of Economics and Management, School of Information Science and Engineering, Yanshan University, Qinhuangdao 066004, China

Corresponding author: Binbin Li (ysulbb@gmail.com)

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ABSTRACT As a universal interaction method, hand and finger gestures can express people's intention directly and clearly in daily life, and has been one of the hotspots in human-computer interaction community. In this paper, we present *DMT*, a device-free finger gesture tracking system that can track and recognize the finger motion accurately. To achieve this, we transform the mobile device, such as a smart phone, into an active sonar system by establishing inaudible audio links between the built-in speakers and microphone. The finger motion will have an effect (e.g., Doppler-shift) on the audio signals, which makes it possible to track the finger motion according to the received signal characteristics at the microphone side. Due to the small reflection energy and slow moving speed of the finger, the existing methods cannot detect the Doppler-shift accurately. To this end, a Fourier fitting-based method is proposed in the *DMT* to accurately detect the Doppler-shift. With the detected Doppler-shift, the *DMT* can track the finger motion with high accuracy. The *DMT* supports all kinds of finger gestures interaction, including characters and shapes. Extensive experiments demonstrate the high accuracy and robustness of the *DMT* in dynamic environments.

INDEX TERMS Device free, finger gesture tracking, Fourier fitting, Doppler-shift.

I. INTRODUCTION

A. BACKGROUND

The past decade has witnessed the explosive growth of various mobile devices, including smart phone, PAD, smart watch, wristband, etc [1]. Due to the small dimension of operation interface on these devices, the human-computer interaction technology for the purpose of intelligent control has been one of the research hotspots [2]. Different from traditional keyboard-based and mouse-based methods, the intelligent control of mobile devices requires higher interaction friendliness and convenience [1]. In all the interaction technologies, the gesture-based methods may be the best choice because of its strong naturalness, simplicity, richness, and substantivity.

B. EXISTING WORKS AND LIMITATIONS

Existing gesture tracking and recognition methods for mobile devices can be classified into four categories, namely wearable devices based methods [3]–[6], vision and images based methods [7], [8], RF signal based methods [9]–[12] and acoustic signal based methods [13], [14]. The wearable

devices based methods [3], [4] require the operator to wear sensor-augmented devices, and recognize the gestures according to the sensor data. While these methods are intrusive detection [15] and not friendly to use in practice because most of them affect the daily life of operator. The vision and images based methods require extra deployment of cameras and other devices [7], which will also lead to a leakage of other private information. Moreover, the changes of environment such as light and smoke may also have a great impact on the recognition accuracy. Although RF signal based methods can overcome most of aforementioned limitations, they are extremely sensitive to the environment changes, any change of the environment will lead to a decline of recognition accuracy. And it is impossible to guarantee a stable RF signal coverage in some places, such as on the buses, subways, trains, etc., which leads to the unavailability of these methods. As the most of Commercial-Off-The-Shelf (COTS) mobile devices have built-in speaker (transmitter) and microphone (receiver), many acoustic signal based methods have been proposed in recent years [15]. They can work well on various applications and do not require any extra equipment. While most of them

suffer from the problem of rigorous gesture mode requirements, such as the gesture amplitude [10] and speed [14] should be large enough.

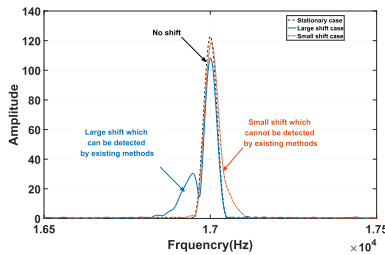


FIGURE 1. Different doppler-shifts caused by different gestures.

C. MOTIVATION

In practice, the amplitude and velocity of different gestures have great diversity, especially for different operators. Obviously, a large gesture motion will cause significant effect (e.g., doppler-shift) to acoustic signal, which makes it easy to be tracked. While the effect of a micro-gesture is so weak that it is challenge to track it. Figure 1 shows three doppler-shift examples caused by different gestures when the source acoustic frequency is set to be 17kHz. From the figure, we can get two key observation. First, the frequency distribution of the received acoustic signal has significant symmetry at the center of the source frequency in the stationary case when there is no finger motion, which has not been well used in previous works. Second, the doppler-shift caused by a large gesture is also large, the shift peak (the second peak) can be clearly observed, and can be easily detected by existing methods [16]. While the doppler-shift caused by a micro-gesture is small as shown by the orange curve, the shift peak is not apparent, and cannot be detected by existing methods. All these motive us to design a much more general scheme for the purpose of device-free micro-gesture tracking.

D. MAIN CONTRIBUTIONS

We propose a Device-free Micro-gesture Tracking method called *DMT* in this paper. *DMT* establishes audio communication links between the built-in speakers (transmitters) and microphone (receiver) using acoustic side channels. Based on the symmetry observation, *DMT* employ a fourier series fitting method to get the exact doppler-shift value. With the detected doppler-shift during the finger motion, *DMT* tries to track and recognize the finger micro-gestures accurately. There are several challenges to achieve this. First, because of the small reflection area, the doppler-shift caused by the finger's micro-gesture is so weak that it is difficult to detect with existing doppler-shift detection methods. Second, it is also not so easy to establish the relationship between the detected doppler-shift and the finger gestures.

The main contributions of this paper can be summarized as follows:

- We propose a device-free micro-gesture tracking scheme called *DMT* for mobile devices. Using the acoustic side channels, *DMT* establishes audio communication links between the built-in speakers and microphone in mobile devices, and does not require any extra equipment.
- Based on the symmetry of FFT result, we design a novel fourier series fitting based doppler-shift detection method to accurately detect the doppler-shift during the gesture motion.
- We also design a geometric method to track the trajectory of gesture with detected doppler-shift, and identify the gesture via an image matching algorithm.
- We implement a prototype system of *DMT* on COTS smart phone, and evaluate its performance via extensive experiments. The results demonstrate that *DMT* can track and recognize the finger motion with more than 95% accuracy.

II. RELATED WORK

In the past years, with the emergence and explosive growth of mobile devices (e.g., smart phones), human-computer interaction for mobile devices has be one of the hotspots in both research and industrial communities. In all these interaction technologies, gesture tracking and recognition has received more attention because of its naturalness and directness. A number of gesture tracking and recognition solutions have been proposed and greatly promote the development of human-computer interaction. Existing gesture tracking and recognition methods can be classified into the following four categories:

A. WEARABLE DEVICES BASED GESTURE RECOGNITION

These methods are mainly based on various wearable sensors or devices which should be carried by the gesture operator, such as smart headband [17], [18], digital glove [19], smart wrist [20] and smart watch [21]. By collecting data from the sensors (i.e. accelerometer), a mapping model between the sensory data and gesture motion is established to recognize the operator's gestures. There are also some studies which require the operators to wear special wearable devices. For example, Ketabdar *et al.* [22] designed a wearable magnet device and implemented a virtual signatures system for mobile devices. Lu *et al.* [6] designed a prototype device embedded with a three-axis accelerometer and four surface electromyographic (SEMG) sensors, and a bayes linear classifier was used to identify 19 predefined gestures.

B. VISION-BASED GESTURE RECOGNITION

Because of the disadvantages of wearable sensor-based methods, such as invasive and expensive, the non-invasive methods based on computer vision have received a lot of attention from research community in recent years. Such vision-based methods mainly utilize the captured images from the camera, extract the gestures from the images using image processing approaches, and then used the methods like Hidden Markov Models (HMM, [23]), Conditional Random Fields

(CRF, [24]), Support Vector Machines (SVM, [25]), Deep Neural Networks (DNN, [26]) to classify and recognize the 2D [27] or 3D [28] gestures.

C. RF-BASED GESTURE RECOGNITION

The emergence of wireless network has also greatly promoted the research and application of RF-based gesture recognition. According to different RF sources, these RF-based methods can be classified into two categories, namely RFID-based and WiFi-based methods. By attaching a cheap RFID tag on the finger, Wang *et al.* [29] designed an antenna array consisted of eight antennas in subtle ways, and used the Angle of Arrival (AoA) of RF signal to locate the finger's position, and recognized the gestures via trajectory matching. On the contrary, RF-Care [30] tried to get the gestures via tag array. Alqaness and Li [10] designed a fast Dynamic Time Warping (DTW) algorithm to identify different gestures with the received WiFi signals. Adib *et al.* [11], [12] proposed another WiFi-based method called Wivi and even can detect the gesture motion behind wall. The core idea of Wisee [9] is that the gesture motion will cause different doppler-shift in different OFDM channels. Using the directional antennas, Melgarejo *et al.* [31] can recognize more than 25 different gestures. Kellogg *et al.* [32] proposed AllSee, a novel gesture-recognition system that can be used for computing devices, no matter how low-end and power-constrained they are. Taking advantages of the detailed channel state information (CSI) available from commodity WiFi devices, Tan and Yang [33] proposed WiFinger which can sense and identify subtle movements of finger gestures by examining the unique patterns exhibited in the detailed CSI.

D. SONAR-BASED GESTURE RECOGNITION

All the above-mentioned methods require extra deployments, and so they are not very convenient in practical applications. On the other hand, there are many built-in sensors (e.g., microphone, camera) on COTS mobile devices. So it is a good idea to implement the gesture tracking using these built-in sensors. Based on the intuitive understanding of the gesture motion must change the acoustic signals between speaker and microphone, lots of sonar-based gesture tracking methods have been proposed. SonarGest [34] was the earliest acoustic-based hand gesture recognition system. The core idea of SonarGest is to utilize one ultrasonic sender and three receivers, calculate the feature vectors of different hand gestures using the supervised Gaussian Mixture Model (GMM), and identify different gestures according to distributions of feature vectors. Nandakumar *et al.* [35] proposed a contactless solution for detecting sleep apnea events on smart phones. They transform the phone into an active sonar system that emits frequency-modulated sound signals and records their reflections. Similarly, FingerIO [13] can track human finger motions, and an intrusion detection solution is proposed in [36]. Both SoundWave [37] and AudioGest [1] utilized the doppler-shift of acoustic signals to recognize the gesture without any training in advance.

III. PRELIMINARIES

A. ACOUSTIC SIDE CHANNEL

As we know, the general frequency bandwidth of human voice is 50HZ~11kHz, and the bandwidth of human hearing is 20HZ~20kHz. To support various audio-based applications, such as speech interaction, it is necessary for COTS mobile devices to cover the whole frequency bandwidth from 20HZ to 20kHz. So all the COTS mobile devices can emit and record sound waves with frequency lower than 20 kHz. Actually, most of COTS mobile devices cut off the acoustic signal which above 20kHz directly by using a Low Pass Filter (LPF).

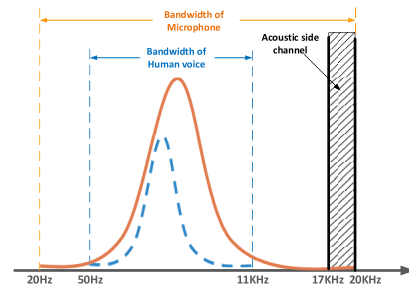


FIGURE 2. Acoustic side channel.

Figure 2 shows the relationship of these bandwidths. In the whole bandwidth from 20HZ to 20kHz, there is a special bandwidth from 17kHz to 20kHz, which is only partly audible to infants and pets, and inaudible to most people. This means that this bandwidth is rarely used for COTS mobile devices. We call this bandwidth from 17kHz to 20kHz as acoustic side channels in this paper.

The usage of acoustic side channel has several advantages. First, it is usually rarely used, so it does not bring any negative effects to other applications on mobile devices. Second, it is inaudible to most people, so it also does not cause any discomfort to anyone. Third, the frequency is high and the wavelength is short, so any small movement of gesture will cause significant effect to the acoustic signals.

B. DOPPLER EFFECT AND DOPPLER-SHIFT

The well known doppler effect and doppler-shift have been widely used in so many wireless applications, including Radar [38], WiFi [10], RFID [39] and sonar [13] systems. The propagation speed of sound is much slower than electromagnetic waves, so that the acoustic doppler-shift is usually much more notable. As shown in Figure 3(a), when the finger moves toward the speaker, the microphone will receive a signal with a lower frequency than the frequency of original signal. Otherwise, if the finger moves toward the microphone, the microphone will receive a higher frequency signal. The general equation of doppler-shift measurement is as follows:

$$\Delta f = f_{receiver} - f_{source} = \frac{\Delta v}{v_{wave}} f_{source} \quad (1)$$

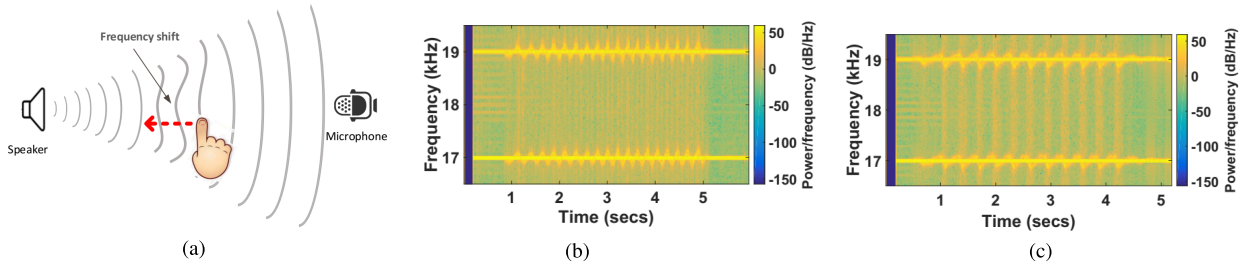


FIGURE 3. Doppler effect and doppler-shift of different gestures. (a) Doppler shift. (b) Round-trip line motions. (c) Circle motions.

where Δf is the doppler-shift, Δv is the velocity of the receiver relative to the source, and v_{wave} is the propagation speed of acoustic signal.

In our case, the mobile device is placed on the table, hence both the source (i.e., speaker) and the receiver (i.e., microphone) are stationary, and only the reflector (i.e., finger) is moving. So the doppler-shift can be measured as follows,

$$\Delta f = \frac{2v_{rad}}{v_{wave}} f_{source} \quad (2)$$

where v_{rad} means the radial speed of finger to microphone.

Figure 3(b) and (c) show the doppler-shift diagrams of different gestures. We can see clearly from these figures that different gestures will cause different doppler-shift features. So it is possible and feasible to identify the gestures via accurate doppler-shift detection.

IV. DMT DESIGN

We give out the detail design of *DMT* in this section.

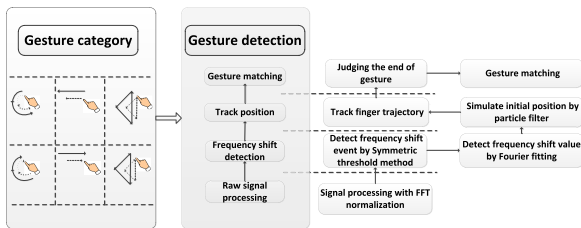


FIGURE 4. System architecture of *DMT*.

A. SYSTEM OVERVIEW

To achieve the objective of device-free micro-gesture tracking, *DMT* consists of four steps, as shown in Figure 4. In the first step of signal preprocessing step, we introduce a FFT-Normalization based method to filter out the influence of audible noise without adding an extra signal filter. Then *DMT* adopts threshold and fourier fitting based methods to accurately detect the doppler-shift in the second shift detection step. And in the gesture tracking step, we utilize the particle filters to estimate the initial position of gesture and then track the finger's position with the detected doppler-shift. At last, *DMT* matches the tracked gesture trajectory with the predefined gestures to determine the final output gesture.

B. SIGNAL PREPROCESSING

Because of the wide bandwidth, the microphone on COTS mobile devices can record acoustic signals from 20HZ to 20kHz. So there must be audible noise in the raw data recorded by microphone. Moreover, the hardware diversity and the temporal changes also may influence the quality of raw data [1], [16].

To eliminate the influence of these noises, we introduce FFT-based normalization in *DMT*, so that an extra band-pass filter is not required. As shown in Figure 5, we first adopt a 1920 points hamming window to segment the raw signal into frames, then apply a 1920 points FFT for each frame to get the frequency domain of acoustic.

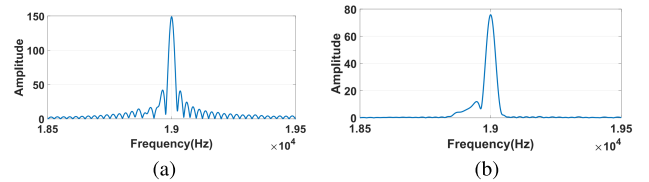


FIGURE 5. FFT results of preprocessing. (a) FFT result without Hamming window. (b) FFT result with Hamming window.

As shown in the figure, there are serious spectrum leakage in the result of FFT without hamming window. With hamming window, we can get a high-quality FFT result and the second frequency peak which is caused by finger motion can be seen clearly in Figure 5(b).

C. DOPPLER-SHIFT DETECTION

So far, we have got a time series spectrum after the signal preprocessing step, next we will show the method to detect the doppler-shift accurately. However, it is not so easy to detect the doppler-shift accurately when only micro-gesture is happened, because finger's reflected energy is so weak. There are two challenges we should well resolved in *DMT*.

1) DOPPLER-SHIFT EVENT DETECTION

The first challenge is the audio signal drift, which can be caused by two reasons, namely the temporal signal drifts and diverse-device signal drifts [1], [16]. The emitted audio signals of the same mobile phone at different time depict various magnitudes, and the different mobile phones also

emit various magnitudes signals. The primary reason of this phenomenon may be the heat effect of electronic components which caused by the continuous high-frequency vibrations, and hence there is no any regular pattern at all, which leads to the useless of traditional energy-based methods.

Fortunately, there are other features which can be used to detect the occurrence of doppler-shift event. As aforementioned, regardless the audio signal drifts are, there is obvious symmetry of FFT results at the center of source frequency in the stationary case. When there is finger motion, the symmetry no longer be valid, so we can use this feature to accurately detect the start and end of a doppler-shift event.

Assuming that people's gestures have a maximum speed of 4m/s [14] and the source frequency is set to be 17kHz, the maximum frequency doppler-shift is about 400HZ, which can be calculated using Equation 2. That means the shift can only present in the bandwidth from 16.6kHz to 17.4kHz. To validate the symmetry, for each FFT frame, we get 400 frequency amplitude on the left and right of 17kHz respectively, and let the right ones subtract the left ones correspondingly, so as to get the amplitude difference of the i th frame as follows:

$$AD_i = FFT_i(right) - FFT_i(left). \quad (3)$$

Figure 6(a) shows the amplitude difference of 5000 frames in the stationary case when there is no gesture motion, and Figure 6(b) shows the one result when there is a finger gesture motion. From the figure, we can get that, the max amplitude difference is so small (i.e., <1) and can be ignore in the stationary case, so the symmetry can be verified. While the symmetry no longer valid when there is gesture motion. So it is possible to distinguish whether there is a gesture motion according to the symmetry of the FFT frames. If the symmetry of FFT frame is valid, then there is no gesture. Otherwise, there should be a gesture motion.

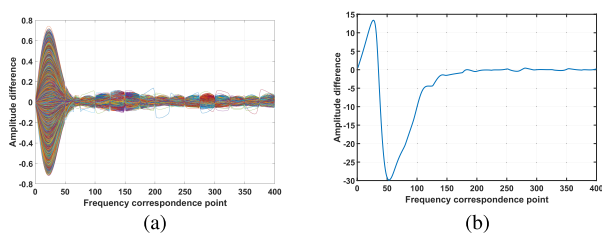


FIGURE 6. Amplitude difference. (a) The symmetry of amplitude difference in stationary case. (b) The asymmetry of amplitude difference in gesture motion case.

Based on the above analysis, we apply a threshold based method to detect whether there is a doppler-shift event in *DMT*. Since the shift may happen at either left side or right side of the center frequency, when the amplitude difference of current FFT frame is larger than the predefined threshold τ , we take this frame as the doppler-shift event start frame, and the corresponding time is the gesture motion start time. Then if the amplitude difference of consecutive five frames is smaller than the τ , we can take corresponding frame and time as the motion end frame and the end time.

The selection of threshold τ is very important, because it will greatly influence the accuracy of doppler-shift event detection. If it is too small, exceptions caused by audio signal drifts will be identified as gesture motion. If it is too large, actual gesture motion will be covered up as hardware noises.

For the purpose of getting an optimal τ value, we first conduct an experiment to get about 2000 FFT frames in static condition, and the cumulative distribution function (CDF) results of amplitude difference are shown in Figure 7.

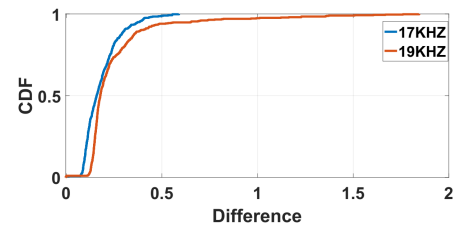


FIGURE 7. CDF of amplitude difference in stationary case.

As shown in the figure, the amplitude differences of 17kHz are much more stable than the differences of 19kHz, all the differences of 17kHz in stationary cases are below the level of 1, and most of differences ($>97\%$) of 19kHz are smaller than 1. So we empirically set the value of threshold τ to be 1 in *DMT*.

2) DOPPLER-SHIFT MEASUREMENT

The second challenge of doppler-shift detection is that, the shift caused by micro-gesture is so weak that it cannot be detected using existing doppler detection methods as aforementioned. Next, we will show a fourier series fitting based method to accurately measure the doppler-shift value, which is also based on our experimental observations about the symmetry of FFT results in frequency domain.

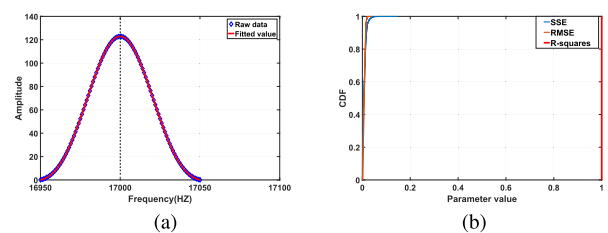


FIGURE 8. Fourier series fitting results with sample data. (a) Fitting results. (b) CDF of fitting results.

We first test and verify the applicability of fourier series fitting in stationary condition. We sample 50 points from the left and right sides of the center frequency respectively, so as to get a sample data set of 101 points, and then utilize the fourier series fitting method to fit the sample data set. Figure 8 shows the fitting results. From the Figure 8(a), we can get that, the output curve fits the sample data well, and both the Sum of Squared Errors (SSE) and Root Mean Squared Errors (RMSE) are close to 0, R-squares are 1 as shown in Figure 8(b). All of these indicates the applicability of the

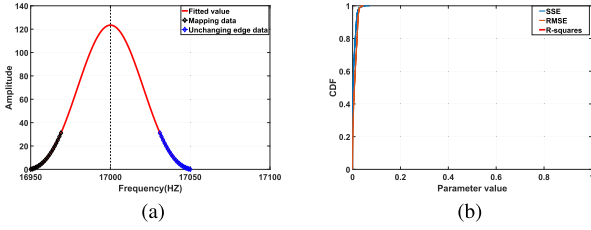


FIGURE 9. Fourier series fitting results with unchanging edge data. (a) Fitting results. (b) CDF of fitting results.

fourier series fitting in stationary condition, next we will show a method to detect doppler-shift value when there is a gesture motion.

However, the fourier series fitting cannot be used in gesture motion condition directly, because it is impossible to get the static component from the FFT results of motion condition via sampling. Fortunately, there is an important feature we can use. The doppler-shift causes the whole band to move to the left or to the right, which means that there must have some edge data which did not affected by the gesture motion at either left side or right side. That is to say, at least the edge data of one side of FFT results in gesture motion condition will highly consistent with the corresponding data in static condition. This feature gives us a chance to get the static component with these unchanging edge data.

The whole procedure to get the static component has three steps, the first step is to find the unchanging edge data, the second is to do a center mapping at the center frequency based on the symmetry, and the last step is to get the fitted curve by using a fourier series fitting method. Figure 9 shows the fitting results with 20 points unchanging edge data of the right side. We can see from the Figure 9(b) that the fitting results also match the original FFT results in static condition well, so the results indicates that in gesture condition, we can utilize unchanging edge data to fit the FFT result of the signal which not affected by the gesture motion and the rest part is the frequency shift signal caused by the gesture.

Then we can turn to get the accurate doppler-shift measurement for each FFT frame in which doppler-shift event has been detected in last subsection. For each FFT frame, *DMT* first finds the unchanging edge data via simply comparing with the edge data from static condition, then using the unchanging edge data, *DMT* try to construct the static component with the fourier series fitting. Finally, *DMT* let the raw data of this frame subtract the static component data, and get doppler-shift from the subtraction results.

Figure 10 shows an example of the above procedure. We can see that, even when the doppler-shift is so weak, and cannot be detected using existing second-peak based doppler detection methods, our *DMT* can detect and measure the doppler-shift accurately. This provides an opportunity for accurate micro-gesture tracking.

Then the parameters of fourier series fitting, including the fitting order and the number of unchanging edge data used in the fitting, also may influence the accuracy of doppler-shift measurement.

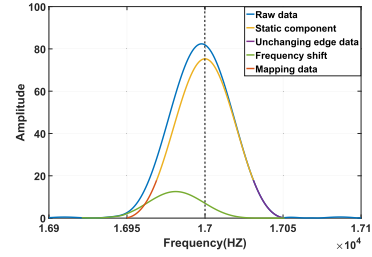


FIGURE 10. An example of doppler-shift measurement.

To get the best parameters, we collected more than 5000 FFT frames in stationary condition, and perform fourier series fitting with different parameters. Figure 11 shows the corresponding results. Comparing with 11 (a) and (b), the RMSE results in 11 (c) are much smaller, which means that the output curves of third-order fitting fit the sample data much better. The increasing of fitting order will synchronously increase the fitting time. So taking the measurement efficiency into consideration, it is sufficient for *DMT* to adopt a third-order fourier series fitting to get the static component with unchanging edge data. Similarly, to the number of unchanging edge data, 20 points unchanging edge data is enough for *DMT* as shown in 11 (c).

D. GESTURE TACKING AND MATCHING

So far, we have got the accurate doppler-shift which is caused by gesture motion. However, one acoustic link between one speaker and one microphone only can produce one doppler-shift, and it is not enough for gesture tracking. Fortunately, most COTS mobile devices have multiple speakers and microphones integrated. Hence, we adopt two speakers and one microphone in our *DMT*, let these speakers emit different acoustic signals in different frequency of 17kHz and 19kHz respectively, and the microphone records all the received signals.

1) TRACKING MODEL

The tracking model used in *DMT* is shown in Figure 12. S_1 and S_2 are the left and right microphones on the mobile devices. Let the position of S_1 as the origin of coordinate, namely its coordinate is (0, 0). And the distance from S_1 to S_2 is D , so the coordinate of S_2 is $(D, 0)$.

Supposing the position of finger at the time of t is (x_t, y_t) , and the distance from finger to S_1 is $D_{t,1}$, the distance from finger to S_2 is $D_{t,2}$, T_s is the time interval of each FFT frames. When the finger moves to a completely new position of (x_{t+1}, y_{t+1}) , and the detected doppler-shifts of two links are $F_{t+1,1}$ and $F_{t+1,2}$ respectively, so the new distance of $D_{t+1,1}$ and $D_{t+1,2}$ can be calculated as follows:

$$D_{t+1,1} = D_{t,1} + \left(\frac{F_{t+1,1}}{F_1} v_{wave} \right) T_s, \quad (4)$$

$$D_{t+1,2} = D_{t,2} + \left(\frac{F_{t+1,2}}{F_2} v_{wave} \right) T_s, \quad (5)$$

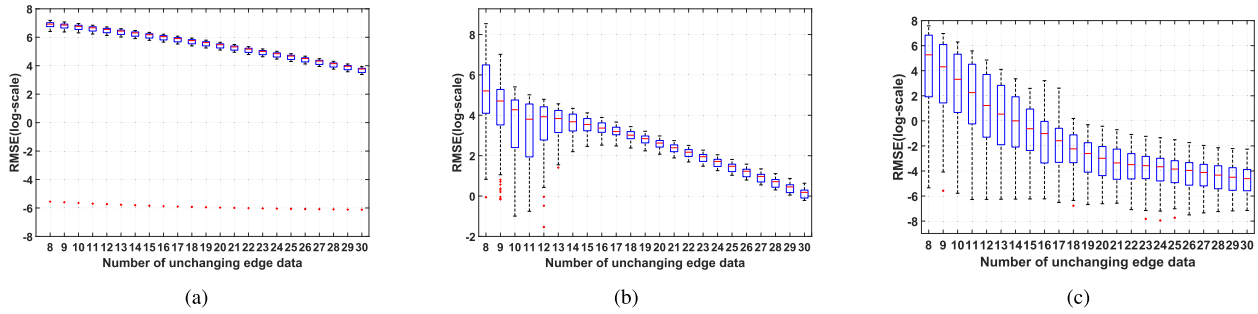


FIGURE 11. Fourier series fitting results with different parameters. (a) Results of first-order fitting. (b) Results of second-order fitting. (c) Results of third-order fitting.

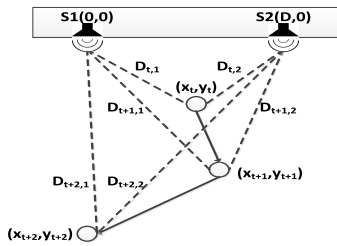


FIGURE 12. The gesture tracking model.

where F_1 represents the frequency of the sound signal emitted by S_1 , and F_2 is the frequency of the sound signal emitted by S_2 , V_{wave} stands for the velocity of sound in the air.

With $D_{t+1,1}$, $D_{t+1,2}$ and the known location of S_1 , S_2 and the position of new point (x_{t+1}, y_{t+1}) forms a triangle with S_1 and S_2 , thus we can get the new position of the finger (x_{t+1}, y_{t+1}) using following equations:

$$\theta = \cos^{-1}\left(\frac{D_{t+1,1}^2 + D^2 + D_{t+1,2}^2}{2DD_{t+1,1}}\right), \quad (6)$$

$$(x_{t+1}, y_{t+1}) = (D_{t+1,1}\cos\theta, D_{t+1,1}\sin\theta). \quad (7)$$

So, we can get the new position of (x_{t+1}, y_{t+1}) with (x_t, y_t) and doppler-shift $D_{t+1,1}$, $D_{t+1,2}$. In the same way, we can get (x_{t+2}, y_{t+2}) on the basis of (x_{t+1}, y_{t+1}) . The procedure can be repeated until a doppler-shift ending event which means the ending of gesture motion is detected. We can get the trajectory of the gesture motion by taking all the positions together.

However, the upon tracking model is based on the hypothesis of knowing the initial position of the gesture, which is impossible to know in advance. To solve the problem of unknown initial position, we introduce particle filter in our *DMT*.

First, we generate a large number of particles in the possible area where gestures may occur, and utilize each particle as a candidate initial position for the gesture. For each detected Doppler shift, we constrain the filtered particles with a triangular law, and the sum of any two sides should be greater than the third side. Any particles that do not meet the constraints

will be filtered out and discarded. So, as the gesture motion progresses, the number of candidate particles will decrease.

Theoretically, the more initial particles, the more opportunity to get the accurate gesture position. On the other hand, the increasing of particles will also cause the increasing execution time of *DMT*. So that a trade-off between accuracy and execution time of *DMT* is needed, and the problem will be resolved in the next evaluation section.

2) GESTURE MATCHING

With the tracked trajectory, we then try to match trajectory with predefined gestures via a image matching. Both the trajectory and predefined gestures are converted to images, denoted as Tr and $\{pg_i | i \in [1, n]\}$ respectively, where n is the number of predefined gestures. The purpose of gesture matching is to find the most similar gesture pg_i for Tr . Because of translation, rotation and scaling, the trajectory Tr may greatly deviate from the predefined gestures. So it is infeasible and inefficient to get the matching result with a direct matching.

To get an accurate matching, we introduce a image matching method from image processing into *DMT*. First, the contours of both tracked trajectory and predefined gestures are detected using the Candy algorithm [40], which is a multi-level edge detection algorithm for detecting the contour of a figure. The contour detection results are represented as $C(Tr)$ and $\{C(pg_i) | i \in [1, n]\}$ respectively.

As a representative feature, the Hu-moment [41] of the same geometric figure will keep invariant, no matter how it is rotated, scaled and translated. Then the Hu-moment features are calculated for each contour figure results, and seven moment features are extracted. The features are denoted as $\{M_1(C(Tr)), M_2(C(Tr)) \cdots M_7(C(Tr))\}$ and $\{M_1(C(pg_i)), M_2(C(pg_i)) \cdots M_7(C(pg_i)) | i \in [1, n]\}$ respectively.

Then the euclidean distance d_i between Hu-moment features of Tr and pg_i are computed. And the k th gesture which satisfies the following equation:

$$k = \arg \min_i d_i, i \in [1, n], \quad (8)$$

is taken as the most similar gesture comparing with Tr . So that the finger gesture is recognized. To enhance the

matching efficiency, the Hu-moment features of predefined gestures can be pre-calculated and stored in advance.

V. EVALUATION

A. EXPERIMENTAL SETTING

Smart phone is a typical mobile devices widely used in daily life, then we implement a prototype of *DMT* on the smart phone. We mainly experiment *DMT* with one smart phone, HuaWei GRA-TL00, which has totally two built-in speakers and two microphones.

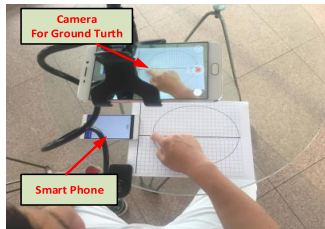


FIGURE 13. The experiment scene.

As shown in Figure 13, the distance between the two speakers on the bottom of the smart phone is 4.4cm, so we let $D = 4.4\text{cm}$. The left speaker emits acoustic signals at 17kHz, and the right speaker emits acoustic signals at 19kHz, the volumes of audio are set to be 50-100 to ensure that it can be recorded clearly by the microphone. To completely understand the performance of *DMT*, we conduct extensive experiments with three different participants (U1, U2, U3 respectively) and three different scenarios, including laboratory (E1), hall (E2), and outdoor (E3). The ground truth is recorded by the camera of another smart phone.

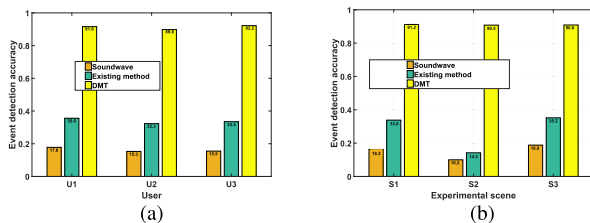


FIGURE 14. Doppler-shift event detection accuracy. (a) With different participants. (b) In different scenarios.

B. DOPPLER-SHIFT EVENT DETECTION ACCURACY

In this experiment, we investigate the doppler-shift event detection accuracy of *DMT*. We let all the three participants to do different finger gestures in the same scenarios of E1. They can do their gestures in their own way, and do not have any requirement about the gestures patterns, including the motion amplitude and speed of the gesture. For each participant, we totally collect acoustic signals of 5 minutes, and there are nearly 3000 FFT frames after preprocessing step. Figure 14(a) shows the accuracy of event detection. We can see from the figure that the accuracy of Soundwave [37] and existing second-peak based method only can achieve event

detection accuracy below 35%, regardless the participants. While the event detection accuracy of *DMT* is close to 90% for all the participants.

Then, we test the event detection accuracy of one participant U2 in different scenarios. The accuracy is shown in Figure 14(b). We also can see from the figure that, Soundwave [37] gets the worst performance with accuracy below 20%, and existing second-peak based method can achieve about 30% accuracy. The reason is that the doppler-shift caused by micro-gesture is too weak to be detected. On the contrary, our *DMT* gets the best performance of more than 90% in all the scenarios, which demonstrates the applicability of *DMT* for device-free micro-gesture tracking in practice.

C. GESTURE TRACKING ACCURACY

Next, we examine the gesture tracking accuracy of *DMT*. The aforementioned tracking model is used, the initial start position of gesture is set to a $20\text{cm} \times 15\text{cm}$ area at the bottom of the smart phone, and particles filter is used to filter out particles in gesture detection process. Taking the time efficiency into consideration, we set the initial number of particles to be 630, which will roughly takes 3.63ms for a gesture tracking.

Figure 15 shows three example tracking trajectories when the participant tries to draw a small circle, a large circle and a triangle respectively. As shown in the figure, although there are some exceptions which significantly deviate from the original shapes, the output shapes drawn by *DMT* are close to the ground truths in most cases and follow the original shapes well. So it is easy to reconstruct the original shapes with the tracked trajectories.

D. GESTURE MATCHING ACCURACY

In this section, we will show the impact of other parameters settings on the final gesture matching accuracy.

Figure 16(a) shows the matching accuracy of different participants in different experimental scenarios. We can see from the figure that, all the accuracy are more than 80%, and achieve about 87% in average. In all the three scenarios, the accuracy of E1 is the lowest, the reason can be the multipathes of laboratory environment is much more complex than other two scenarios, which results a decrease of the matching accuracy. Moreover, the matching accuracy of the third participant U3 is slightly higher than other participants, the reason may be that U3 is much fatter than others, and hence his finger reflects much more acoustic signal during the gesture motion.

Figure 16(b) shows the impact of different table material on matching accuracy. We totally experiment three kinds of table, including wooden table (T1), plastic table (T2) and glass table (T3). From the figure, we can get that the material of table does not have any influence on the matching accuracy at all. The behind reason is that *DMT* is based on the doppler-shift of frequency domain rather than RSSI or phases of acoustic signals, so the influence of table material only impacts the signal propagation but does not impact the doppler-shift values.

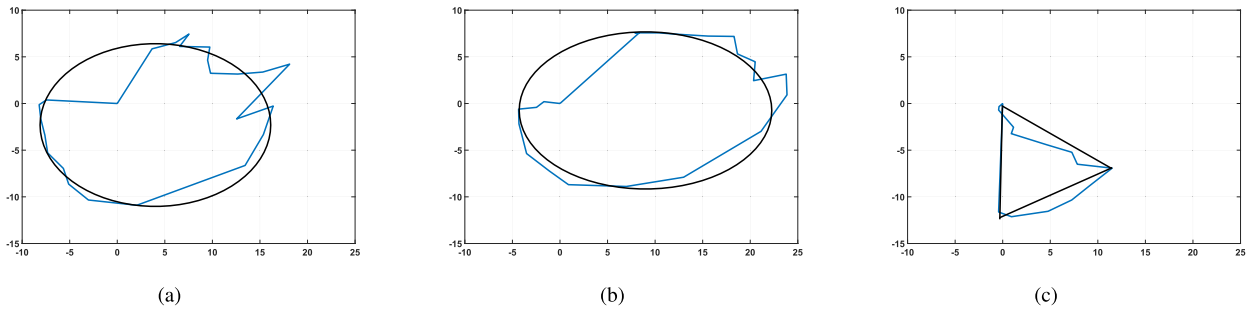


FIGURE 15. Gesture tracking accuracy. (a) Small circle motion. (b) Large circle motion. (c) Small triangle motion.

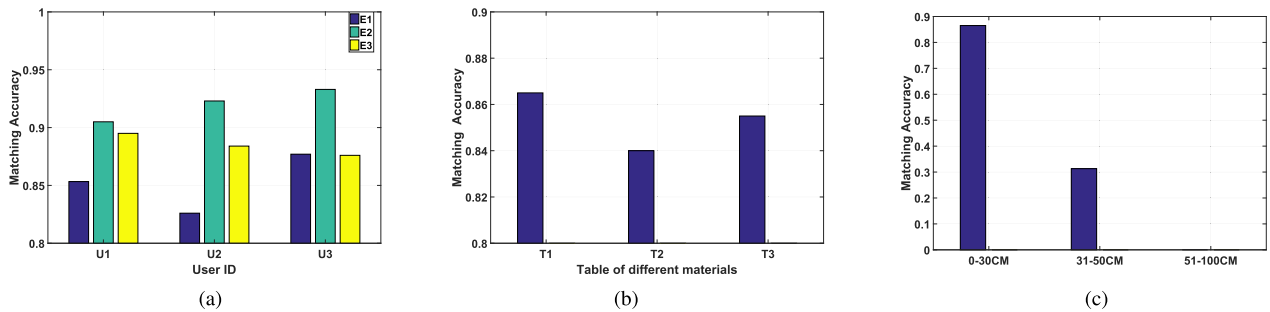


FIGURE 16. Gesture matching accuracy under different settings. (a) In different scenarios. (b) On different material tables. (c) With different distance.

Next, we evaluate the impact of distance from the bottom of smart phone to finger on the matching accuracy. Figure 16(c) shows the experimental results, when the distance is increasing from 30cm to 100cm. We can see from the figure that, there are significant performance declines with the increasing of distance. The matching accuracy within 30cm is acceptable and larger than 89% in average. However, when the distance is larger than 50cm, the matching accuracy is close to 0, which means that *DMT* can not detect any gesture and fail to tracking the motion. It suggests that it is better for the user to do their gesture close to the mobile devices for the purpose of accurate gesture tracking.

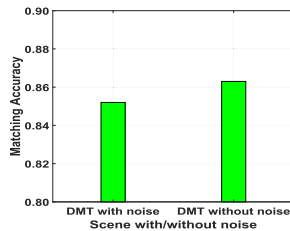


FIGURE 17. Gesture matching accuracy with/without noise in E1.

We then perform additional experiments in a noise settings in a laboratory (E1), where ten people keep talking and walking. Meanwhile, there are also two smart phones which playing music with different volume. Figure 17 shows the robustness of *DMT* against background noise. From the figure, we can see that, *DMT* still can achieve more than %85 matching accuracy within substantial interference.

The reason why the performance with noise is slightly smaller than the one without background noise is the participant may walking too close to the gesture operator. Therefore, the background noises do not affect the performance of *DMT* in most cases, as *DMT* utilizes the acoustic side channel more than 17kHz, and the frequency of human voice and most of music do not exceed 10kHz.

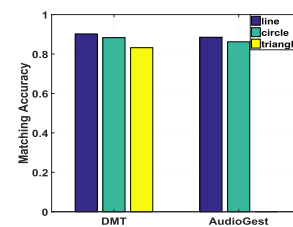


FIGURE 18. Matching accuracy comparison between *DMT* and *AudioGest*.

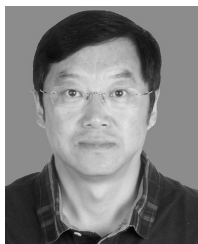
In the end, we compare *DMT* with the latest and representative doppler-shift based solution *AudioGest* [16] in terms of gesture matching accuracy. Figure 18 shows the comparison results. We can see from the figure that, our *DMT* can get almost same matching accuracy as *AudioGest* when the line and circle gesture are performed in a large amplitude. While for the micro-gesture triangle which is performed in a range of 10cm, the *AudioGest* even fails to detect the gesture motion. And our *DMT* still can achieve more than %82 matching accuracy for such micro-gestures. This suggests that our *DMT* is much general and universal for practical gesture tracking applications.

VI. CONCLUSION

In this paper, we propose a novel doppler-shift detection based micro-gesture tracking system called *DMT*. Comparing with existing solutions, *DMT* does not require any extra equipment. Leveraging the built-in speakers and microphones on COTS mobile devices, *DMT* establishes communication links using acoustic side channels, and tracks the gesture motion with the doppler-shift of acoustic signal, which can be detected accurately using a fourier fitting method. Our extensive experiment results demonstrate the feasibility and effectiveness of our *DMT*, which makes an forward step towards enabling accurate and ubiquitous gesture tracking.

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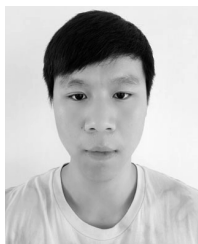


wireless sensor networks and mobile networks.

WENYUAN LIU received the B.S. and M.S. degrees from the Northeast Heavy Machinery Institute, China, and the Ph.D. degree from the Harbin Institute of Technology, in 2000, all in computer science. Since 1996, he has been with the School of Information Science and Engineering, Yanshan University, Qinhuangdao, China, where he is currently a Professor. He is also a special Invited Researcher with the IoT of TNLlist, Tsinghua University. His research interests include



BINBIN LI received the B.E. and M.E. degree from the School of Computer and Technology, Harbin Institute of Technology, China. He is currently pursuing the Ph.D. degree with the School of Information Science and Engineering, Yanshan University, China. He is currently an Assistant Professor with the School of Economics and Management, Yanshan University. His research interests include mobile networks and RFID.



WEIHANG SHEN received the B.S. degree in software engineering from the Hebei University of North China Institute of Aerospace Engineering, Langfang, China, in 2016. He is currently pursuing the M.S. degree in the software engineering with Yanshan University. His research interests include mobile networks and mobile computing.



LIN WANG received the B.Ed. degree from Northwest Normal University and the M.E. degree in computer engineering from Yanshan University. He has been a Visiting Student and a Research Assistant with the GreenOrbs Group, since 2010. He is currently an Associate Professor with the School of Information Science and Engineering, Yanshan University. His research interests include sensor and wireless networks and mobile computing.

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